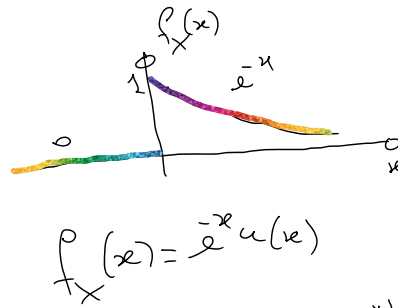


X, Y espressioni

$$E[X] = 1 \quad E[Y] = \frac{1}{2}$$



$$Y = aX \quad a = \frac{1}{2}$$

$$E[Y] = aE[X] = a$$

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y}{a}\right)$$

$$= 2 f_X(2y)$$

$$Z = B(1-X) + (1-B)(Y+2)$$

$$B \in \{0, 1\} \text{ indep. di } X \text{ e } Y \quad P[B=1] = \lambda$$

$$Z|_{B=0} = Y+2$$

$$f_{Y+2}(y) = f_Y(y-2) = 2e^{-2(y-2)} u(y-2)$$

$$= \begin{cases} 0 & y < 2 \\ 2e^{-2y} & y > 2 \end{cases}$$

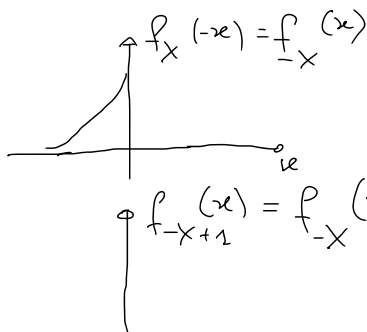
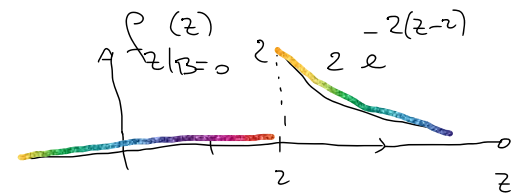
$$Z|_{B=1} = 1-X$$

$$Y = X+a$$

$$= \begin{cases} 0 & y < 2 \\ 2e^{-2(y-2)} & y > 2 \end{cases}$$

$$f_{X+a}(x) = f_X(x-a)$$

$$f_{-X}(x) = f_X(-x)$$



$$f_{-X+1}(x) = f_{-X}(x) = f_X(-x) = f_X(-x+1)$$

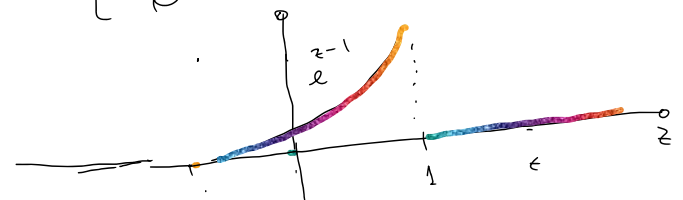
$$= f_{-X}(x-1) = f_X(-(x-1)) = e^{-(-(x-1))} u(-(x-1))$$

$$f_{Y+2|B=0}(z)$$

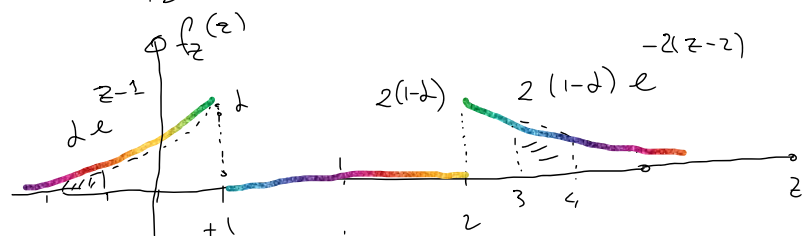
$$f_{Z|B=0}(z) = f_{Y+2}(z) = 2e^{-2z} u(z)$$

$$f_{Z|B=1}(z) = f_{-X+1}(z) = e^{-(z-1)} u(z-1)$$

$$= \begin{cases} e^{x-1} & x < +1 \\ 0 & x > +1 \end{cases}$$



$$f_Z(z) = P[B=0] f_{Z|B=0}(z) + P[B=1] f_{Z|B=1}(z) = (1-\lambda) 2e^{-2z} u(z) + \lambda e^{-(z-1)} u(z-1)$$



$$0 < Z < 4$$

$$P[3 < Z < 4]$$



$$P[3 < |Z| < 4] = P[3 < Z < 4] + P[-4 < Z < -3] = \int_3^4 f_Z(z) dz + \int_{-4}^{-3} f_Z(z) dz$$

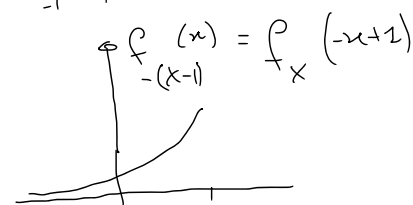
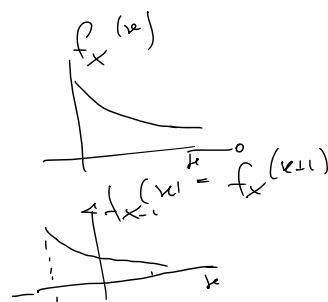
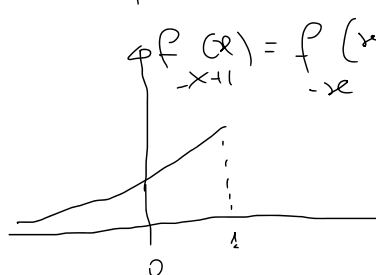
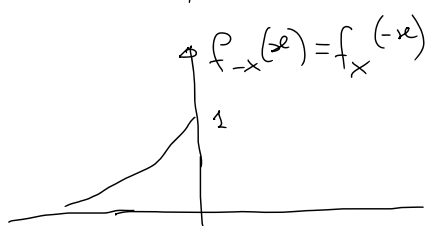
$$3 < |Z| < 4 \Leftrightarrow (3 < Z < 4) \text{ OR } (-4 < Z < -3)$$

$$E[Z] = \int_{-\infty}^{+\infty} z f_Z(z) dz$$

$$= P[B=0] E[Z|B=0] + P[B=1] E[Z|B=1] = (1-\alpha) \frac{5}{2}$$

$$E[Z|B=0] = E[Y+Z|B=0] = E[Y+2] = E[Y] + 2 = \frac{5}{2}$$

$$E[Z|B=1] = E[1-X|B=1] = E[1-X] = 1 - E[X] = 0$$



$$\int_{-\infty}^{+\infty} z f_Z(z) dz = \int_{-\infty}^{+\infty} z \left[\alpha e^{z-1} u(1-z) + (1-\alpha) z e^{-2(z-2)} u(z-2) \right] dz$$

$$= \int_{-\infty}^1 z \alpha e^{z-1} dz + \int_2^{+\infty} (1-\alpha) z z e^{-2(z-2)} dz$$

$$= \alpha \left\{ \left[z e^{z-1} \right]_{-\infty}^1 - \int_{-\infty}^1 (1 \cdot e^{z-1}) dz \right\} + (1-\alpha) \left\{ \left[z \left(-\frac{1}{2} e^{-2(z-2)} \right) \right]_2^{+\infty} - \int_2^{+\infty} \left(-\frac{1}{2} e^{-2(z-2)} \right) dz \right\}$$

$$= \alpha \left\{ 1 - \left[e^{z-1} \right]_{-\infty}^1 \right\} + (1-\alpha) \left\{ 2 + \left[-\frac{e^{-2(z-2)}}{2} \right]_2^{+\infty} \right\} = F(b) - F(a)$$

$$= 2 \left\{ \cancel{1} - (1) \right\} + (1-2) \left\{ 2 + \frac{1}{2} \right\} = \frac{5}{2}(1-2)$$

$$P[A] \cdot f_{X/A}(x) = P[A|X=x] f_X(x)$$

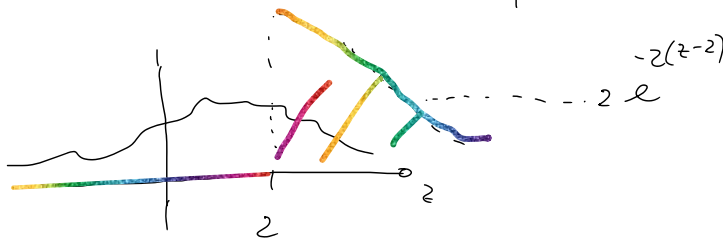
$$E[Z | Z > 2]$$

$$f_{Z|Z>2}(z) = \frac{P[Z > 2 | Z=z] f_Z(z)}{P[Z > 2]} = \begin{cases} 0 & z < 2 \\ \frac{f_Z(z)}{P[Z > 2]} & z > 2 \end{cases}$$

$$P[Z > 2 | Z=z] = P[Z > 2] = \begin{cases} 1 & z > 2 \\ 0 & z < 2 \end{cases}$$

$$P[Z > 2] = \int_2^{+\infty} f_Z(z) dz$$

$$\frac{f_Z(z)}{P(Z > 2)} = \frac{1}{\int_2^{+\infty} f_Z(z) dz} f_Z(z)$$



$$P(Z > 2) = \int_2^{+\infty} f_Z(z) dz = \int_2^{+\infty} \left[2 e^{z-1} u(1-z) + (1-2) z e^{-2(z-2)} u(z-2) \right] dz$$

$$= \int_2^{+\infty} (1-2) z e^{-2(z-2)} dz = (1-2) \left[-\frac{1}{2} e^{-2(z-2)} \right]_2^{+\infty} = (1-2) \times 1$$

$$E[Z | Z > 2] = \int_{-\infty}^{+\infty} z f_{Z|Z>2}(z) dz = \int_{-\infty}^{+\infty} z \frac{f_Z(z)}{P[Z > 2]} u(z-2) dz$$

$$= \frac{1}{P(Z > 2)} \int_2^{+\infty} z (1-2) z e^{-2(z-2)} dz = \int_2^{+\infty} z^2 e^{-2(z-2)} dz = \frac{5}{2}$$

$$Z = B(1-X) + (1-B)(1+2)$$

$$\rightarrow \quad \rightarrow \quad z_1 \quad - \quad |2+1| \quad = 2+1$$

$$Z = 10 \cdot \dots \cdot \dots \cdot \dots \cdot \dots$$

$$Z > 2 \Leftrightarrow B = 0 \quad Z|_{Z>2} = Z|_{B=0} = (2+Y)|_{B=0} = 2+Y$$

$$E[Z|Z>2] = E[2+Y] = 2 + \frac{1}{2} = \frac{5}{2}$$

$$Z = BX + (1-B)Y$$

$$\phi_Z(s) = E[e^{-sZ}] \quad s > 0$$

$$E[g(Z)] = \int_{-\infty}^{+\infty} g(z) f_Z(z) dz = \int_{-\infty}^{+\infty} g(z) \left[P(B=0) f_{Z|B=0}(z) + P(B=1) f_{Z|B=1}(z) \right] dz$$

$$g(x) = e^{-sx} \quad = P(B=0) \left(\int_{-\infty}^{+\infty} g(z) f_{Z|B=0}(z) dz \right) + P(B=1) \left(\int_{-\infty}^{+\infty} g(z) f_{Z|B=1}(z) dz \right)$$

$$E[g(Z)] = P[B=0] E[g(Z)|B=0] + P[B=1] E[g(Z)|B=1]$$

$$Z|_{B=0} = Y|_{B=0} = Y \Leftrightarrow g(Z)|_{B=0} = g(Y)$$

$$Z|_{B=1} = X|_{B=1} = X \Leftrightarrow g(Z)|_{B=1} = g(X)$$

$$E[g(Z)|B=0] = E[g(Y)] \quad E[g(Z)|B=1] = g(X)$$

$$E[g(Z)] = (1-\alpha) E[g(Y)] + \alpha E[g(X)]$$